DD2356 – More on Cache Memory

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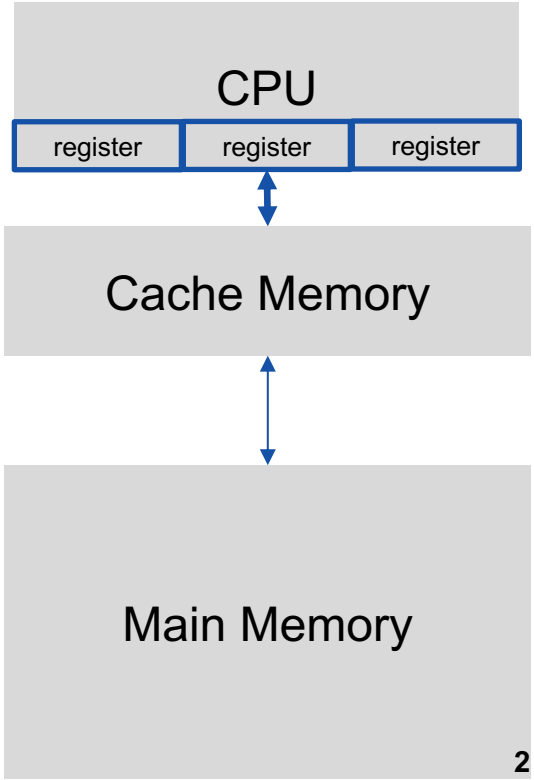
Simplified Computer Architecture

- Main memory contains the program data
  - **Cache memory contains a *copy* of the main memory data**
    - Cache is *faster* but consumes more space and power
    - Cache items accessed by their address in main memory
- **Registers contain working data only**
  - Modern CPUs perform most or all operations only on data in register

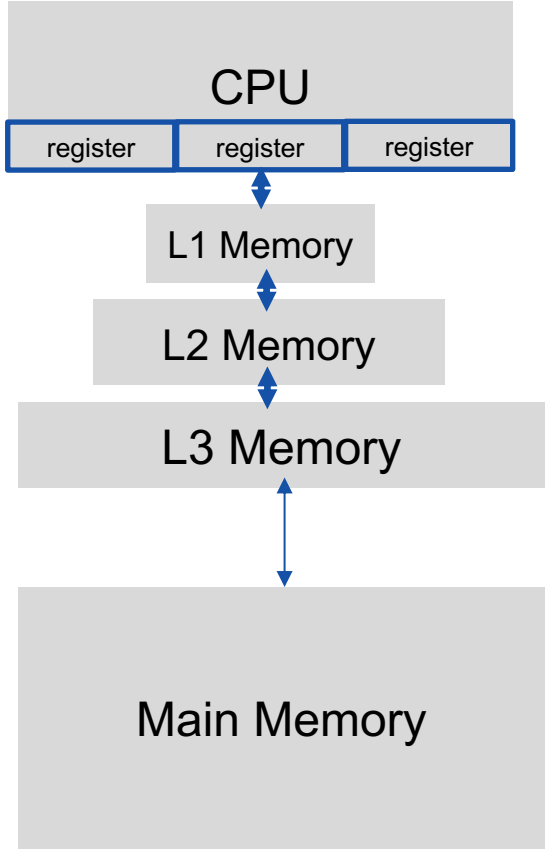


Importance of Memory in Performance Bounds

- We have seen:
  - **Loads and stores can be as important as floating-point operations**
  - Simple models that look at just sustained memory bandwidth (and ignore details of cache effects) can provide useful *bounds* on performance
    - Recall the sparse matrix-multiply example
      - True for problems where the majority of data accesses are consecutive
  - Note that this is a bound, a guaranteed-not-to-exceed value for the performance

Simplified Computer Architecture II

- Because of the way cache is implemented in hardware, there are tradeoffs between
  - Size (number of bytes or capacity)
  - Speed
  - Power
- **L1 cache is the fastest** but has the least capacity
- **L2** and **L3** (also LLC = last level cache) **provide intermediate performance/size tradeoffs**

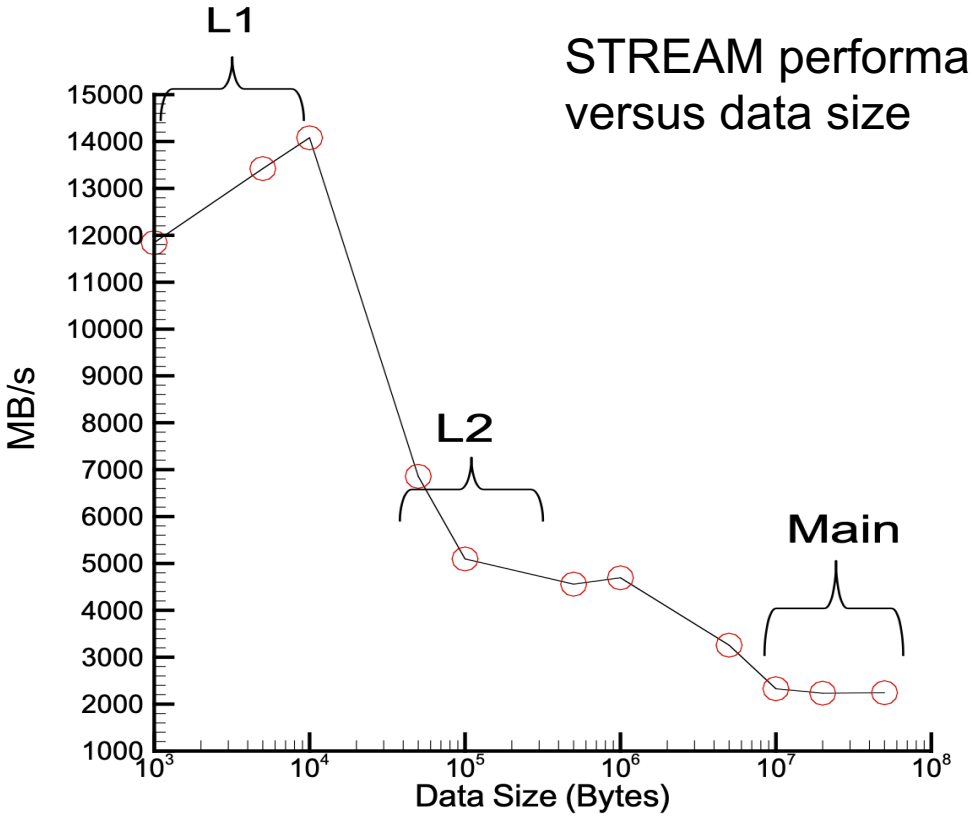


Memory Locality

Typical access times for cache-based systems

Register	1 cycle	Ratio to previous
L1 Cache	Few cycles	**1-3x**
L2 Cache	~ 10 cycles	3-10x
Memory (DRAM)	~ 250 cycles	**25x**
Remote Memory (Out of Node)	~2500-5000 cycles	**10-20x**

Impact of Memory Hierarchy



STREAM performance in MB/s versus data size

Updating the Sparse Matrix Example

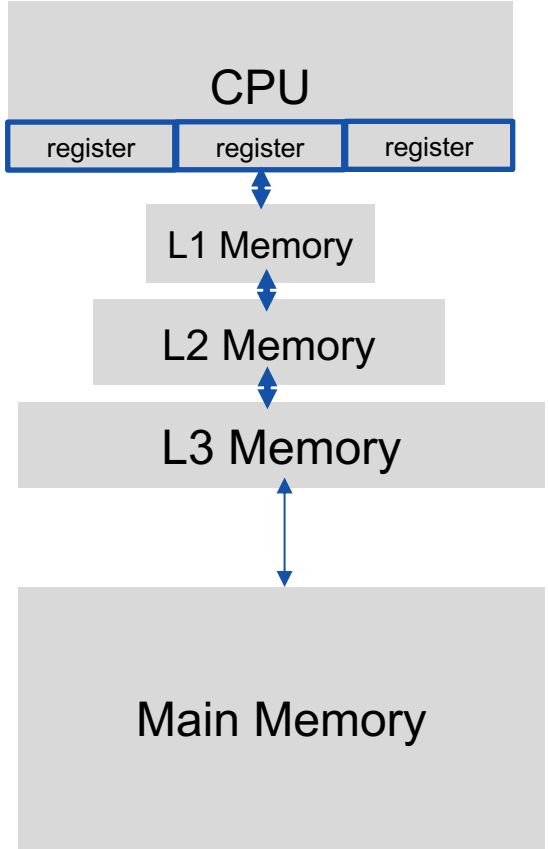
- Original (no cache)
  - `Time = nnz(2c + 2.5r) + n_row(0.5r+w)`
- Add in time to load `x` from Cache:
  - `Time = nnz(2c + 1.5r +` r_c`) + n_row(1.5r +w)`
    - Where r_c is the time to read from (the appropriate level) of cache
    - Note that `x` is read into cache from memory (the `1` in the `n_row(1.5r)` term)

More on Cache Operation

- Most communication in a computer is carried out in **chunks**
  - blocks of bytes of data that move together
- In the memory hierarchy, data moves
  - between memory and cache, and
  - between different levels of cache
- in groups called ***cache lines***

Simplified Computer Architecture III

- Main memory contains the program data
  - Multiple cache memories contain a copy of the main memory data
- Data is moved between levels of memory in **groups of words called lines**
  - Lines are typically **64-128 bytes**,
    - 8-16 double precision words
  - Even if you don't use the data, it is moved and occupies space in the cache
- Programming Language Relevance
  - **This performance feature is not captured in most programming languages** ☹



Types of Locality

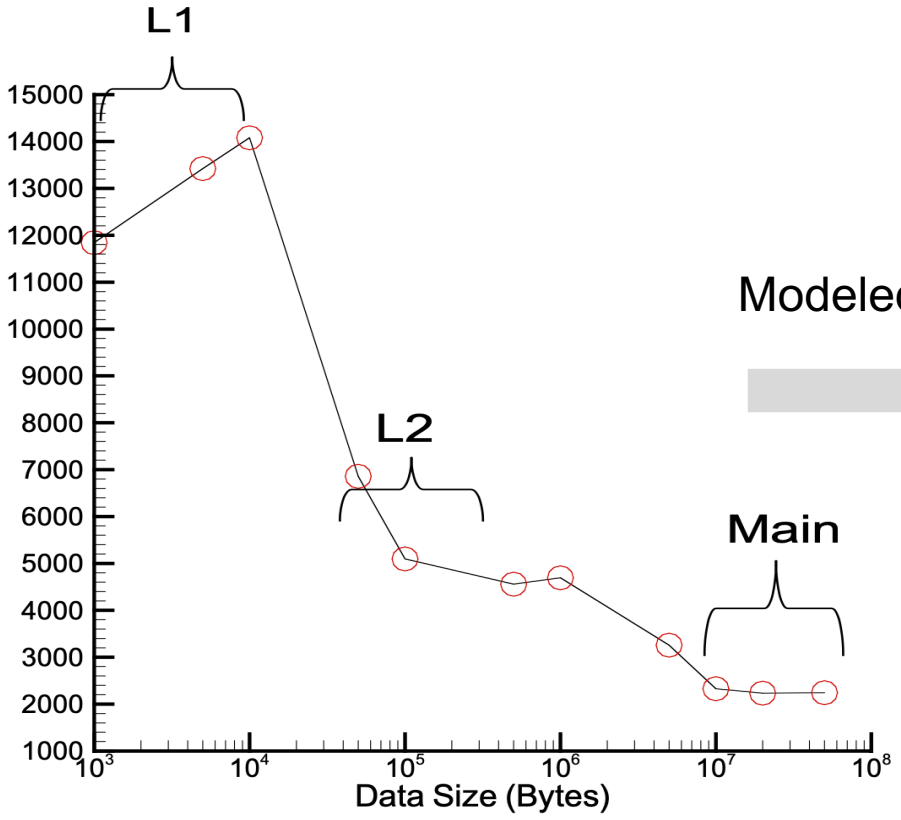
- **Temporal – reuse same data**
  - This has been our assumption in the sparse matrix case – data is reused
- **Spatial** – (re)use “nearby” data
- These match the *hardware*
- Alpern and Carter * suggest a different breakdown, based on needs of algorithms:
  - **Local:** Data has both Temporal and Spatial
  - **Semi-local:** Data has Spatial locality
  - **Nonlocal:** Data has neither spatial nor temporal locality

* Performance Programming for Scientific Computing, SIAM Course

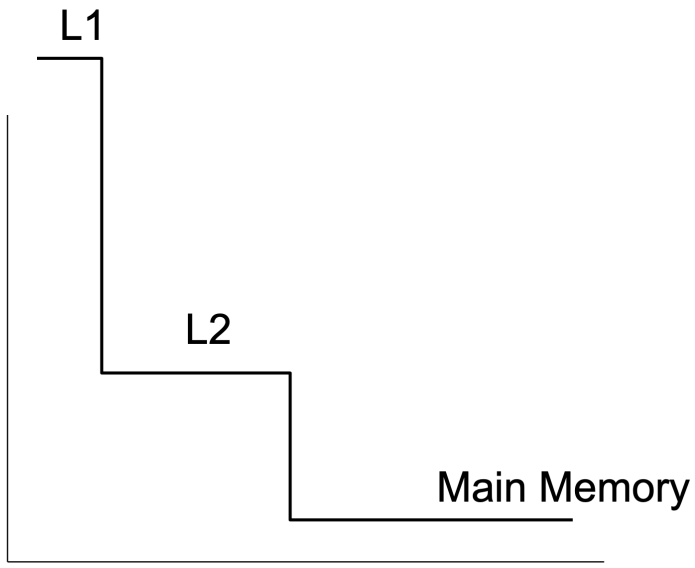
Refining the Bounds

- **Achievable memory bandwidth depends on small, fast, cache memory**
  - *Temporal locality* reduces need for sustained memory bandwidth
    - Data is present in faster memory
- Leads to an important rule-of-thumb in performance modeling
  - Identify dominant terms
  - Architectures are too complex for any other approach
  - Similar to approaches for mathematical modeling
- "Does it fit in cache" provides a first model
  - Assume all of cache is available
  - For multiple cache levels, **use performance of the smallest (fastest) cache** in which the data fits

Impact of Memory Hierarchy: Measured / Modeled



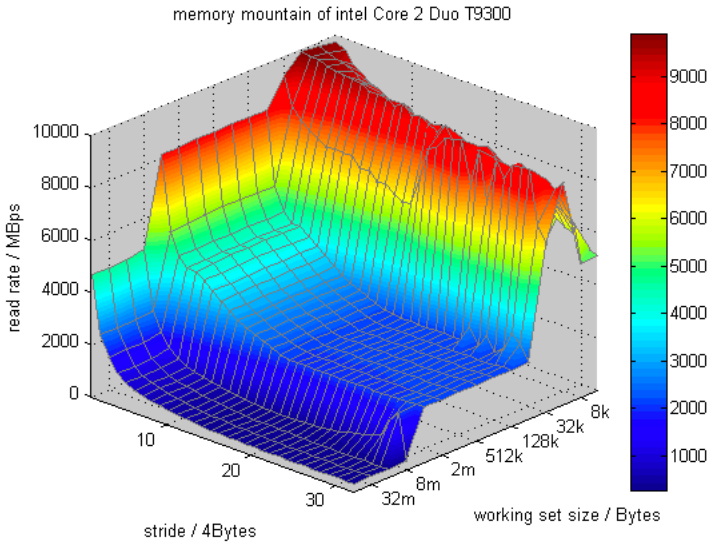
Modeled



Refining the Bounds: Spatial Locality

- Non-consecutive memory accesses expose more details about the memory structure
  - Effective cache size reduces when not all of the data on the same cache line is used
    - *Spatial Locality* important to get full use of cache
      - **Cache line size** helps in refining performance bounds
        - **Reduce effective cache size to same fraction of cache line used**.
        - E.g., if 1 8 byte value used from a 128-byte cache line, or 1/16 of the line, the effective cache size is 1/16 of the full size

Putting it Together: the Impact of Caches on Program Performance

- **Memory Mountain** – A two-dimensional function of load (read) bandwidth versus temporal and spatial locality
- Characterizes the capabilities of the memory system for each computer

- One of your assignments